2024-DSE
PHY
PAPER 1B

B

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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2024

Candidate Number

PHYSICS PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Graph paper and supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Short Questions Question No.	Marks
1	6
2	6
3	6
4	6
5	4
6	5
7	6
8	6
9	6
Long Questions Question No.	**Marks**
10	10
11	7
12	8
13	8

A150E01B

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81$ m s^{-2}.

1. A bottle of water of temperature 4 °C is taken out from a refrigerator and placed on a table. The room temperature is 30 °C.

Figure 1.1

(a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)

(b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)

(c) After a period of time, water droplets form on the surface of the bottle.

(i) State the source of the water droplets. (1 mark)

(ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)
Given: specific latent heat of vaporization of water = 2.26×10^6 J kg^{-1}

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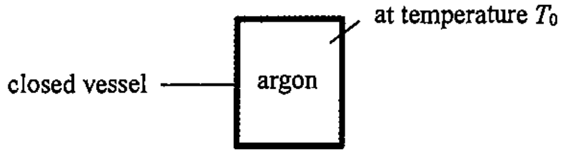
*2. Figure 2.1 shows a closed vessel of a fixed volume containing argon, a monatomic gas, maintained at temperature T_0. The root-mean-square speed ($c_{r.m.s.}$) of the argon atoms inside is 500 m s^{-1}.

Figure 2.1



(a) (i) Calculate the average kinetic energy of the argon atoms, E_K. Given that the mass of an argon atom is 6.63×10^{-26} kg. (2 marks)

(ii) Hence, or otherwise, find T_0 (in K). (2 marks)

(b) If some of the argon has been leaked from the vessel while the temperature remains at T_0, state and explain the change, if any, in the root-mean-square speed of the argon atoms inside the vessel. (2 marks)

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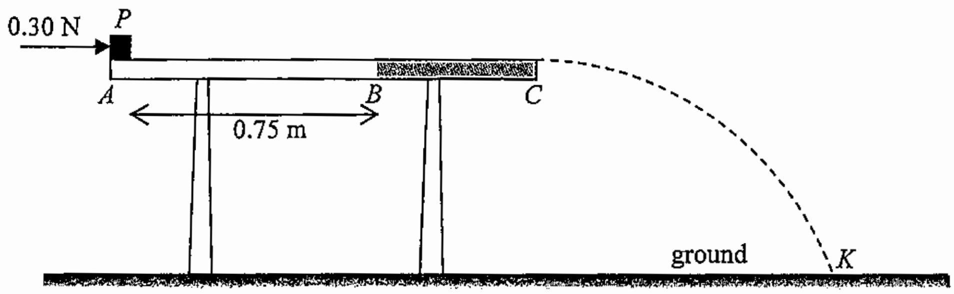
3. A small block P of mass 0.20 kg rests on side A of a horizontal table as shown. The surface of the table is smooth from A to B but is rough from B to C. Neglect air resistance.

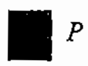
Figure 3.1



A horizontal force of 0.30 N acts on the block as shown.

(a) Find the speed of the block after it travels 0.75 m to point B. (2 marks)

(b) The horizontal force is removed when block P just reaches B. When the block continues to travel from B to C, on the free-body diagram below indicate and name the force(s) acting on the block. (2 marks)



*(c) The block leaves side C of the table and takes 0.35 s to reach point K on the ground. Estimate the height of the table. (2 marks)

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*4. A sphere of radius 0.50 m rotates about a vertical axis ACB with a constant angular speed of 6 rad s^{-1}, with C being the sphere's centre. A small plasticine P of mass 0.020 kg is stuck on the surface of the sphere such that it rotates on a horizontal plane containing C as shown in Figure 4.1(a).

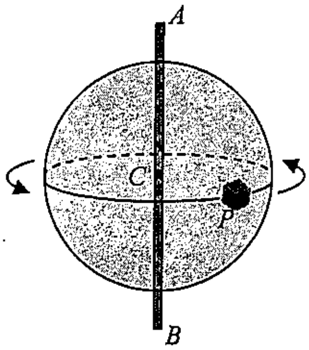

Figure 4.1(a)

Top view from A

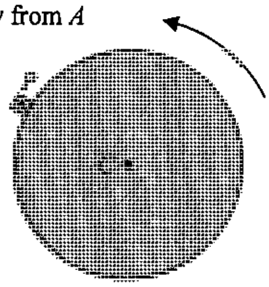

Figure 4.1(b)

(a) (i) Find the magnitude of the centripetal force acting on P. (2 marks)

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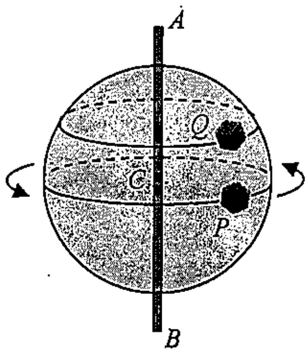
(ii) The sphere's angular speed is increasing. If P finally leaves the sphere at the position shown in Figure 4.1(b) (Top view from A). Indicate on Figure 4.1(b) the direction of motion of P when it **just** leaves the sphere. (1 mark)

(b) Another plasticine Q is stuck on the sphere's surface as shown in Figure 4.2.

Figure 4.2



(i) State whether P and Q have the same angular speed. (1 mark)

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(ii) Explain whether P and Q have the same magnitude of centripetal acceleration. (2 marks)

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5. Figure 5.1 represents the distribution of air particles at a certain instant in a sound wave produced by a loudspeaker.

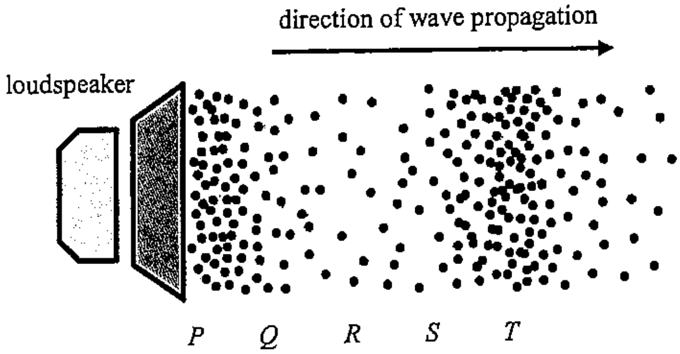
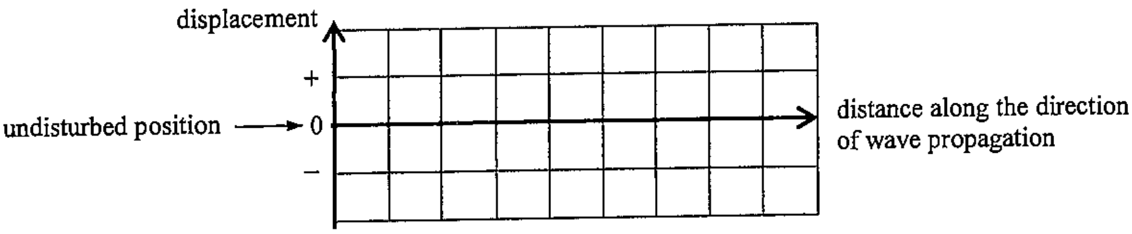


Figure 5.1



(a) Sketch the displacement-distance graph of the air particles at that instant in the grid provided. Take the displacement to the right as positive (+). (2 marks)

(b) At the instant shown, state a region (*P*, *Q*, *R*, *S* or *T*) in which

(i) the air particles are moving with the highest speed; (1 mark)

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(ii) the air pressure is the lowest. (1 mark)

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*6. After removing the protective coating and the reflective layer of a compact disc (CD), the CD becomes a diffraction grating with evenly spaced parallel slits (Figure 6.1(a)). The set-up in Figure 6.1(b) employs a laser beam which can be used to measure the slit separation d.

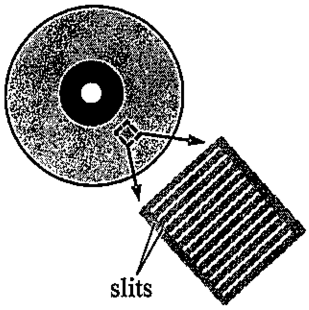

Figure 6.1(a)

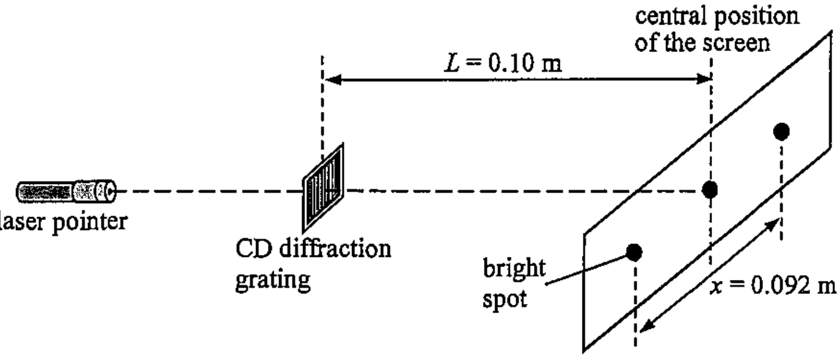

Figure 6.1(b)

Given that the separation between the CD diffraction grating and the screen, L, is 0.10 m, the separation between the first-order maxima, x, is 0.092 m and the wavelength λ of the red laser light beam used is 650 nm.

(a) Estimate d. (3 marks)

(b) State and explain the change in x if green laser light is used. (2 marks)

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7. The graph below shows the speed of sound at different air temperatures.

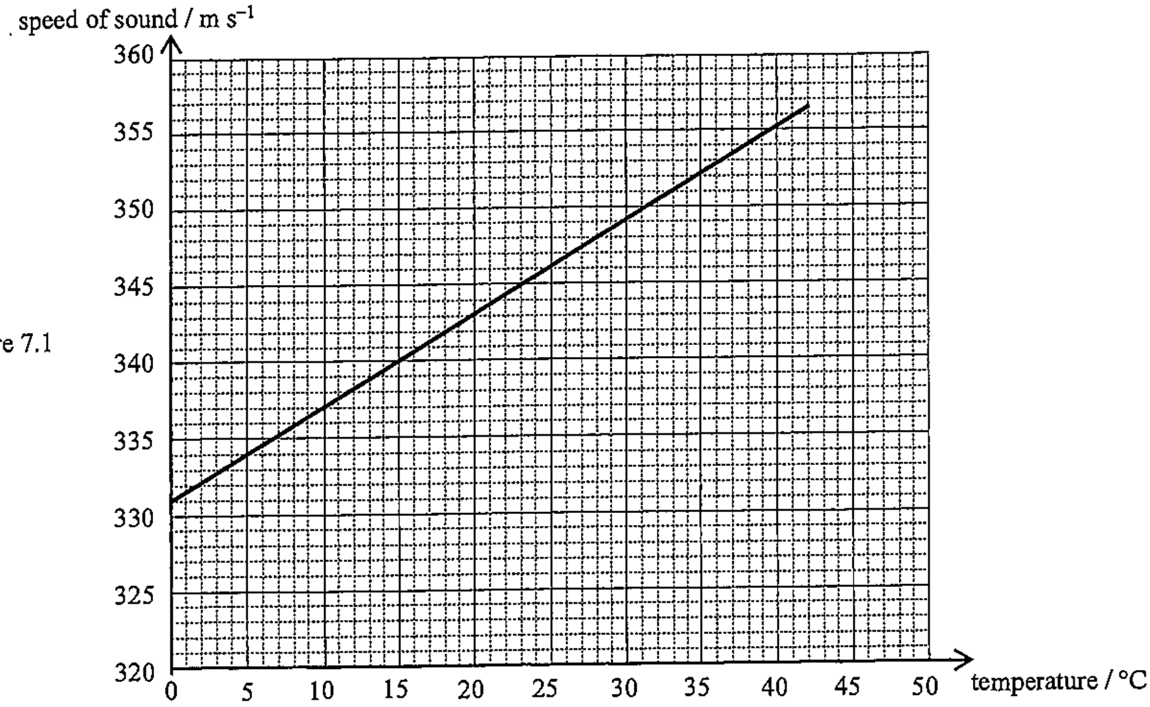

Figure 7.1

(a) With reference to the graph, state the relationship between the speed of sound and the air temperature. (1 mark)

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(b) Figure 7.2 shows sound travelling from a distant tower clock to an observer at night when the ground cools.

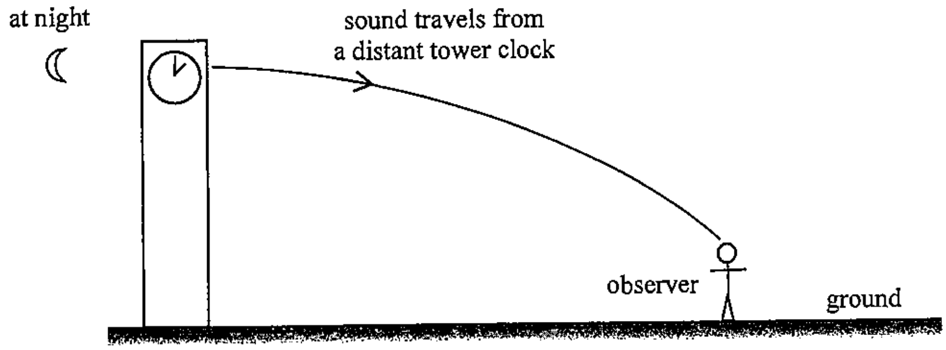

Figure 7.2

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How does the temperature difference between air layers above the ground explain the situation in Figure 7.2 ? (2 marks)

(c) It is known that the air temperature is 10 °C. Referring to the graph in Figure 7.1:

(i) estimate the ratio of speed of light to the speed of sound; (1 mark)

(ii) estimate the distance of a thunderstorm from an observer *O* if thunder is heard 3.0 s after lightning is seen by *O*. (2 marks)

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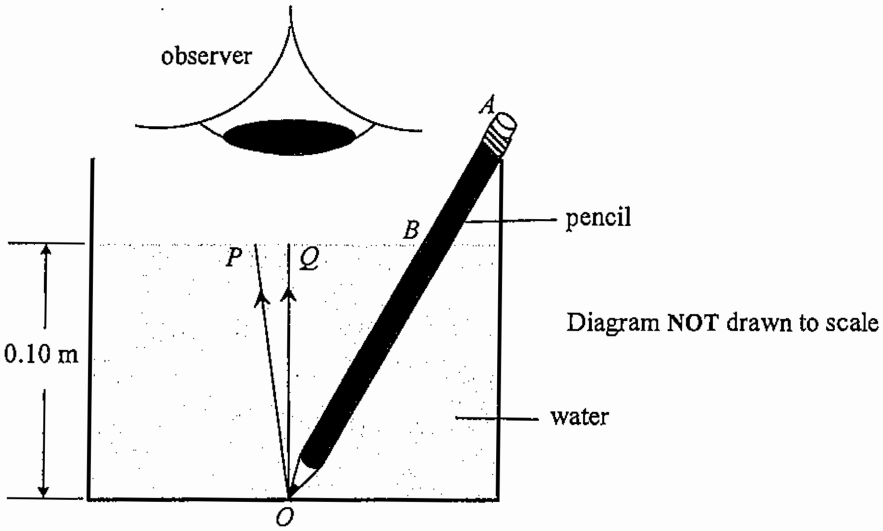
8. Figure 8.1 shows a pencil in a glass of water with an observer viewing vertically downwards. Given that the refractive index of water is 1.33.

Figure 8.1



Light rays P and Q come from the tip O of the pencil, with Q along the vertical.

(a) On Figure 8.1, complete the paths of these light rays and indicate the image of O seen by the observer (label this as I). (2 marks)

(b) The angle between ray P and ray Q is 1.5°. Find the angle of refraction of ray P. (2 marks)

(c) Hence, or otherwise, find the distance between image I and the water surface if the depth of water is 0.10 m. (2 marks)

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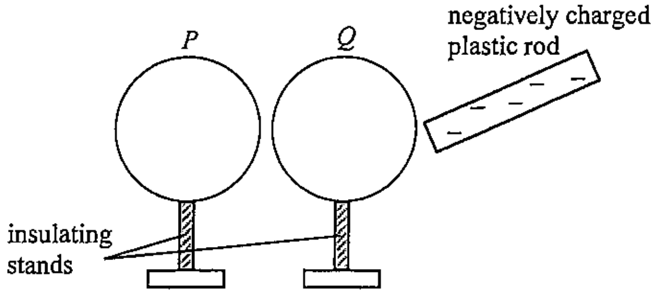
9. Figure 9.1 shows two identical conducting light spheres P and Q placed close to each other without touching.

Figure 9.1



Both spheres are uncharged initially. A negatively charged plastic rod is brought close to Q but without touching the sphere.

(a) On Figure 9.1, sketch the distribution of charges on P and on Q. (2 marks)

A student touches sphere P momentarily with his finger while the plastic rod is in the position as shown. The plastic rod is removed afterwards.

(b) Explain whether the amount of charges on the plastic rod would decrease. (2 marks)

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(c) **The two light spheres** are now suspended by nylon threads from fixed supports such that the spheres are far apart from each other. Explain how you would use one of the spheres to test for the unknown polarity of another charged rod. (2 marks)

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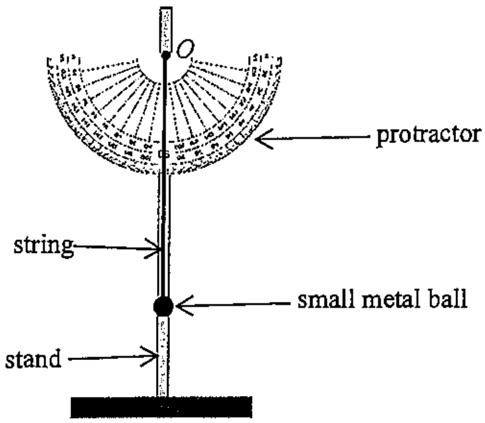
10. (a) As shown in Figure 10.1, one end of an inextensible string is attached to a small metal ball while its upper end O is fixed to a stand. A protractor is fixed onto the stand behind the string.

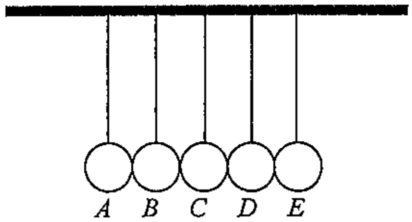
Figure 10.1



Write down the steps that you would take to demonstrate the conservation of energy using the set-up given. (4 marks)

(b) Figure 10.2 shows a Newton's cradle which consists of five identical metal spheres suspended in a row by inextensible light strings of equal length from a fixed horizontal support.

Figure 10.2



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Sphere A and B together are displaced slightly to the left and then released from rest. Just after collision, both A and B come to rest while sphere D and E swing together to the right slightly. Neglect air resistance.

(i) (I) Even though the tensions in the strings acting on the spheres are external forces, explain why the law of conservation of momentum can still be applied for this collision. (1 mark)

(II) Suppose that only one sphere (E) swings to the right with speed v_E just after collision. By finding v_E and hence explain why it is not possible for **only one sphere** to swing to the right. Given that the speed of sphere A and B just before collision is 0.50 m s^{-1}. The mass of each sphere is 0.020 kg. (3 marks)

(ii) The collision repeats alternately from either side for a period of time before the system eventually returns to a rest. Discuss whether the collisions are **perfectly elastic**. (2 marks)

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11. Read the following passage about **deformation sensors** and answer the questions that follow.

Figure 11.1(a) shows a simple sensing device consisting of a thin, flat rectangular metal foil attached to a plastic backing plate. When this foil, length l and width w, is stretched to deform as shown in Figure 11.1(b), its resistance will change.

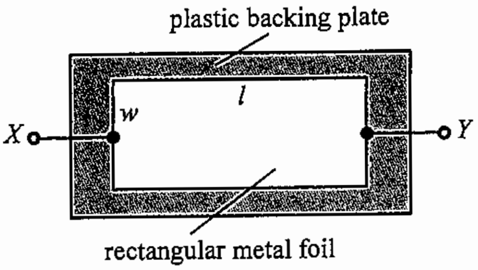

Figure 11.1(a)

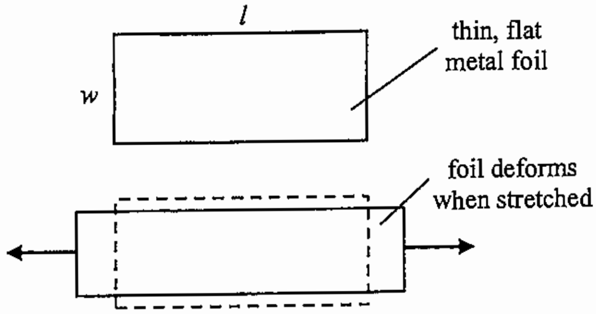

Figure 11.1(b)

When this device is connected to a suitable circuit, it can serve as a sensor to monitor deformation which results in a change in parameters of the circuit (such as voltage).

(a) Referring to Figure 11.1(a) and (b), state and explain how the metal foil's resistance across XY changes when the foil is stretched. Assume that the thickness and resistivity of the foil remain unchanged. (2 marks)

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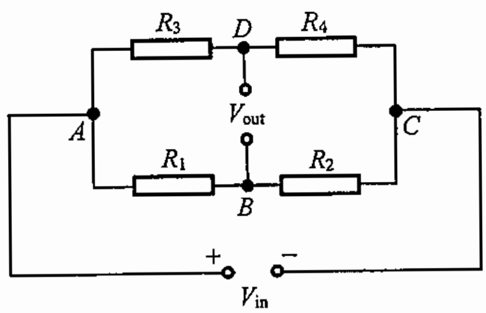
(b) Figure 11.2 shows a circuit with four resistors of resistance R_1, R_2, R_3 and R_4 connected to an input voltage V_{in}. V_{out} is the voltage across DB.

Figure 11.2



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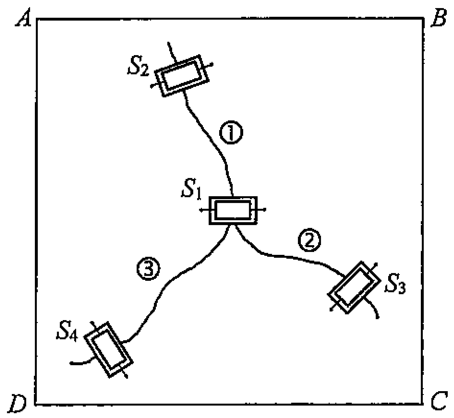
(i) Current I_1 flows through branch ABC. Write an equation relating V_{in}, I_1, R_1 and R_2. (1 mark)

(ii) R_4 in the circuit is the sensing device described in the passage, with unstretched resistance 500 Ω. Find the corresponding percentage change in V_{out}/V_{in} when R_4 increases 0.2% (i.e. becomes 501 Ω).

Given: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$; $R_1 = R_2 = R_3 = 470\ \Omega$. (2 marks)

(iii) Sensing devices S_1, S_2, S_3 and S_4 connected to a suitable circuit are attached to ceiling $ABCD$ as shown in Figure 11.3 in order to monitor the growth of the cracks ①, ② and ③. After a few weeks, the extent of deformation detected is in the order $S_1 > S_2 > S_3 > S_4$.

Figure 11.3



(I) Deduce which crack (①, ② or ③) has grown the most. (1 mark)

(II) Deduce the most probable stretching direction (AB, BC or AC) along the ceiling. (1 mark)

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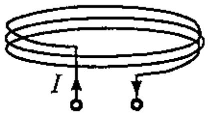
12. Figure 12.1 shows a coil of several turns.

Figure 12.1



(a) Sketch the magnetic field pattern produced by the coil when it carries a current I as shown. (2 marks)

(b) As shown in Figure 12.2(a), this coil is now connected to an a.c. source to serve as a transmitter coil (T). A receiver coil (R) connected to a CRO is placed directly above T. The time variation of the voltage of coil R is displayed on the CRO (Figure 12.2(b)).

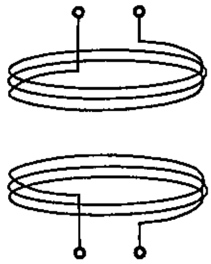


Figure 12.2(a)

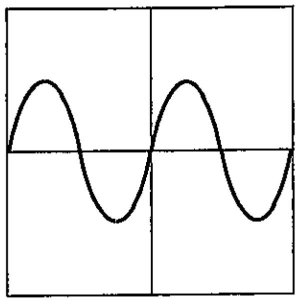


Figure 12.2(b)

(i) Explain how the voltage of the receiver coil R is produced. (2 marks)

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*(ii) On Figure 12.2(b), sketch the CRO display when the frequency of the a.c. source is doubled. (2 marks)

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(c) A mobile phone with a receiver coil inside can be charged wirelessly by placing it on a charging pad equipped with a built-in transmitter coil, which is connected to a power source.

Explain why wirelessly charging of phones need phone cases with non-metallic back covers. (2 marks)

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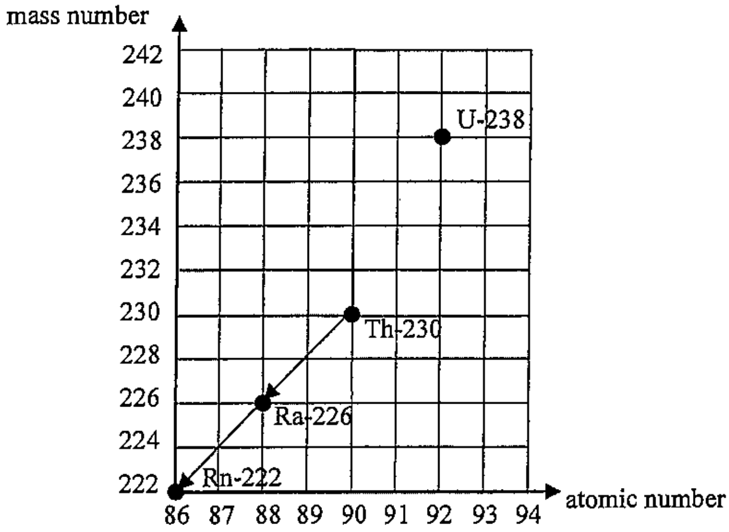
13. Radon (Rn-222) is a radioactive gas with no colour, no smell and no taste. It is released from bedrock material underground or from concrete composed of granite. Rn-222 decays with a half-life of 3.82 days by giving out α particles.

(a) *(i) Find the decay constant (unit: s^{-1}) of Rn-222. (2 marks)

(ii) Inside buildings, the average activity due to radon in 1 m^3 of air is about 48 Bq. Estimate the average number of radon atoms in 1 m^3 of air inside buildings. (2 marks)

(iii) Why is indoor radon considered harmful to human although α particles have limited penetrating power ? (2 marks)

(b) Rn-222 belongs to a decay series starting from uranium-238 (U-238). It is known that U-238 undergoes decay in the sequence α-β-β-α to become thorium-230 (Th-230). Complete the decay series below from U-238 to Th-230 using arrows. (2 marks)



END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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